MAGIVOX AI

GIS Mapping - A to Z

Spatial Data | CRS | QGIS | Digitising | Analysis | Map Design | Interview

How to use this guide
Study one topic, type the example yourself, practise without copying, then take the MAGIVOX topic test. After all modules, take the final course exam.

Learning outcomes

- GIS Foundation
- Vector Data
- Raster Data
- Coordinate Systems
- QGIS Interface
- Loading and Inspecting Data

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1. GIS Foundation

What to learn

- What GIS is
- Spatial vs attribute data
- Real-world GIS use cases
- Vector and raster models

Practice checkpoint

Explain what gis is in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

2. Vector Data

What to learn

- Point, line and polygon
- Attribute table
- Shapefile and GeoPackage
- Topology basics

Practice checkpoint

Explain point, line and polygon in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

3. Raster Data

What to learn

- Pixels/cells
- Resolution
- DEM and imagery
- Raster values and NoData

Practice checkpoint

Explain pixels/cells in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

4. Coordinate Systems

What to learn

- Latitude/longitude
- Geographic vs projected CRS
- EPSG codes
- Why reprojection matters

Practice checkpoint

Explain latitude/longitude in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

5. QGIS Interface

What to learn

- Browser panel
- Layers panel
- Map canvas
- Processing toolbox
- Project save

Practice checkpoint

Explain browser panel in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

6. Loading and Inspecting Data

What to learn

- Add vector/raster/CSV
- Set CRS correctly
- Zoom and identify features
- Filter attributes

Practice checkpoint

Explain add vector/raster/csv in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

7. Digitising

What to learn

- Create layer
- Add fields
- Snapping
- Create/edit features
- Geometry validation

Practice checkpoint

Explain create layer in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

8. Symbology and Labels

What to learn

- Single symbol

- Categorized/graduated
- Label rules
- Legends and readability

Practice checkpoint

Explain single symbol in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

9. Attribute Operations

What to learn

- Field calculator
- Select by expression
- Join tables
- Statistics

Practice checkpoint

Explain field calculator in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

10. Spatial Analysis

What to learn

- Buffer
- Clip
- Intersect
- Dissolve
- Spatial join
- Nearest features

Practice checkpoint

Explain buffer in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

11. Raster Analysis

What to learn

- Hillshade
- Reclassification
- Raster calculator
- Zonal statistics overview

Practice checkpoint

Explain hillshade in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

12. Georeferencing

What to learn

- Control points
- Transformation idea
- Residual error
- Export georeferenced raster

Practice checkpoint

Explain control points in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

13. Map Layout

What to learn

- Title
- Legend
- Scale bar
- North arrow
- Source and CRS
- Export PDF/PNG

Practice checkpoint

Explain title in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

14. Data Quality

What to learn

- Accuracy vs precision
- Missing geometry
- Duplicate features
- CRS mismatch
- Metadata

Practice checkpoint

Explain accuracy vs precision in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

15. GIS Project

What to learn

- Map service coverage or retail locations
- Clean data

- Run analysis
- Create final map
- Explain findings

Practice checkpoint

Explain map service coverage or retail locations in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

16. Interview Preparation

What to learn

- Vector vs raster
- CRS vs projection
- Buffer vs clip
- Spatial join
- Geocoding
- Topology
- Map scale

Practice checkpoint

Explain vector vs raster in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

Worked Examples

Buffer example

```
Create a 1 km buffer around hospitals to study nearby coverage.
```

Spatial join example

```
Count customer points inside each sales territory polygon.
```

CRS example

```
Use a projected CRS for distance/area calculations instead of raw latitude/longitude.
```

Interview Answer Pattern

Use this structure for technical questions:

- 1. Definition - say what it is in one or two lines.
- 2. Why - explain why it is useful.
- 3. Example - give a small real or code example.
- 4. Difference/common mistake - compare with a related concept when relevant.
- 5. Be truthful - do not claim tools or experience you have not used.

Final Course Checklist

- I can explain the foundation without notes.
- I completed examples by typing/doing them myself.
- I completed topic tests and reviewed wrong answers.
- I completed at least one mini project or practical exercise.
- I can answer common interview questions in simple language.
- I am ready for the MAGIVOX 50-question final exam.